



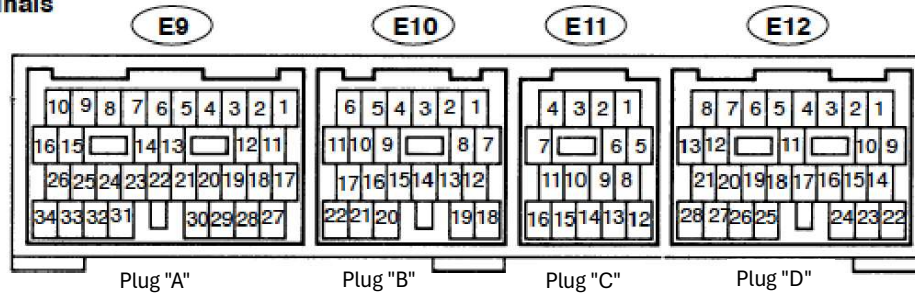
Year  
**4Runner:** 1997-1998  
**Tacoma:** 1998-2000  
**Engine:** 5VZ-FE (3.4L V6)  
**Board:** uaEFI Adapter Board V1.1

#### Notes:

Federal (Non-California) models for 2000 Tacoma

**\*\*WARNING\*\*** Do not plug your ECU into your laptop/computer if your laptop/computer is plugged into the vehicle's 12v charging port!

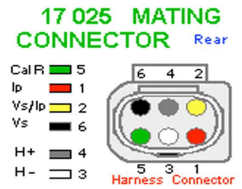
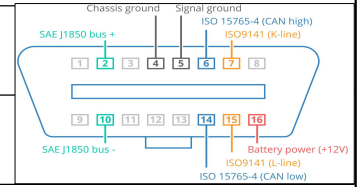
### ECM Terminals



#### OPEN PINS

**Note:** Populate open slots in ECU connectors with MLC040II-SKT1 (OEM p/n 82998-24110) or MLC040II-SKT2 (OEM p/n 82998-24120) pins

| Connector | Plug | Pin | Function                  | Tuner Studio Channel Name | Notes                                                                                                           |
|-----------|------|-----|---------------------------|---------------------------|-----------------------------------------------------------------------------------------------------------------|
| E9        | A    | 3   | Lowside Output (with PWM) | B17 Low Side Output 3     | PWM Ground Output                                                                                               |
| E9        | A    | 4   | 7th Injector              | B9 MR Weak LS 1           | Connect 7th injector negative here, and positive to +B (12V battery)                                            |
| E9        | A    | 18  | Ignition Output 4         | B11 Coil 4                | <b>**MUST be used with an igniter or smart coils**</b>                                                          |
| E9        | A    | 19  | Ignition Output 5         | B13 Coil 5                | <b>**MUST be used with an igniter or smart coils**</b>                                                          |
| E9        | A    | 20  | Ignition Output 6         | B10 Coil 6                | <b>**MUST be used with an igniter or smart coils**</b>                                                          |
| E9        | A    | 21  | AUX2                      | C3 AUX2                   | Analog Input #2 (0-5v)                                                                                          |
| E9        | A    | 25  | CAN1H                     | N/A                       | CANBUS High<br>If connecting to stock OBD2 port, use pin p/n BD-SKT (OEM p/n 776001-1) and populate OBD2 pin #6 |
| E9        | A    | 26  | CAN1L                     | N/A                       | CANBUS Low<br>If connecting to stock OBD2 port, use pin p/n BD-SKT (OEM p/n 776001-1) and populate OBD2 pin #14 |
| E9        | A    | 29  | TPS2                      | C14 TPS2                  | Throttle Position Sensor 2 (Can be used for any 0-5v Analog Input)                                              |
| E9        | A    | 32  | AUX1                      | D1 AUX1                   | Analog Input #1 (0-5v)                                                                                          |
| E10       | B    | 2   | HB1+                      |                           | H-Bridge 1 +                                                                                                    |
| E10       | B    | 3   | HB1-                      |                           | H-Bridge 1 -                                                                                                    |
| E10       | B    | 10  | Sensor Ground             | N/A                       | For sensor grounds only                                                                                         |
| E10       | B    | 11  | G-                        | N/A                       | For separated negative VR leads in the factory wiring harness. Connect camshaft VR- here.                       |
| E10       | B    | 16  | Sensor Ground             | N/A                       | For sensor grounds only                                                                                         |
| E11       | C    | 1   | Flex                      | D5 FLEX                   | Flex Fuel Signal                                                                                                |
| E11       | C    | 4   | 5V Sensor                 | N/A                       | +5V for sensors                                                                                                 |
| E11       | C    | 15  | 5V Sensor                 | N/A                       | +5V for sensors                                                                                                 |
| E12       | D    | 3   | IP                        | N/A                       | For Bosch 17025 (LSU 4.9) WBO (Connector Pin 1)                                                                 |
| E12       | D    | 4   | H-                        | N/A                       | For Bosch 17025 (LSU 4.9) WBO Heater - (Connector Pin 3)                                                        |
| E12       | D    | 9   | VS/IP                     | N/A                       | For Bosch 17025 (LSU 4.9) WBO (Connector Pin 2)                                                                 |
| E12       | D    | 13  | Lowside Output (On/Off)   | MEGA_EPIC_D46             | On/Off Ground Output                                                                                            |
| E12       | D    | 17  | MAP                       | D9 MAP                    | Manifold Absolute Pressure Sensor (Can be used for any 0-5v Analog Input)                                       |
| E12       | D    | 18  | H+                        | N/A                       | For Bosch 17025 (LSU 4.9) WBO Heater + (Connector Pin 4)                                                        |
| E12       | D    | 19  | +B                        | N/A                       | Switched Battery Voltage from EFI Relay                                                                         |
| E12       | D    | 20  | VS                        | N/A                       | For Bosch 17025 (LSU 4.9) WBO (Connector Pin 6)                                                                 |
| E12       | D    | 21  | PPS1                      | D6 PPS1                   | Pedal Position Sensor 1 (Can be used for any 0-5v Analog Input)                                                 |
| E12       | D    | 25  | CalR                      | N/A                       | For Bosch 17025 (LSU 4.9) WBO (Connector Pin 5)                                                                 |
| E12       | D    | 26  | Digital Input             | MEGA_EPIC_D30             | Ground switched                                                                                                 |
| E12       | D    | 27  | Digital Input             | MEGA_EPIC_D31             | Ground switched                                                                                                 |
| E12       | D    | 28  | PPS2                      | C4 PPS2                   | Pedal Position Sensor 2 (Can be used for any 0-5v Analog Input)                                                 |



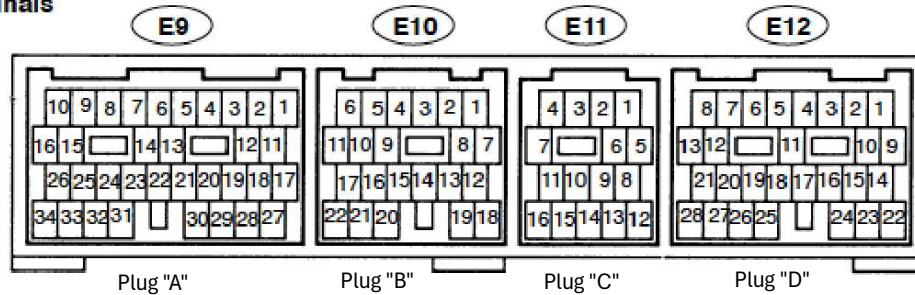


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## POPULATED PINS

| Note: Populate open slots in ECU connectors with MLC040II-SKT1 (OEM p/n 82998-24110) or MLC040II-SKT2 (OEM p/n 82998-24120) pins |      |     |           |                           |                                                                                            |
|----------------------------------------------------------------------------------------------------------------------------------|------|-----|-----------|---------------------------|--------------------------------------------------------------------------------------------|
| Connector                                                                                                                        | Plug | Pin | Function  | Tuner Studio Channel Name | Notes                                                                                      |
| E9                                                                                                                               | A    | 1   | 4WD/AWD   | MEGA_EPIC_D29             | 4WD Indicator                                                                              |
| E9                                                                                                                               | A    | 2   | L4        | MEGA_EPIC_D25             | Transfer Case Low Range                                                                    |
| E9                                                                                                                               | A    | 5   | #60       | B1 Injector Output 6      | Injector #6                                                                                |
| E9                                                                                                                               | A    | 6   | #50       | B2 Injector Output 2      | Injector #5                                                                                |
| E9                                                                                                                               | A    | 7   | #40       | B3 Injector Output 4      | Injector #4                                                                                |
| E9                                                                                                                               | A    | 8   | #30       | B4 Injector Output 3      | Injector #3                                                                                |
| E9                                                                                                                               | A    | 9   | #20       | B5 Injector Output 2      | Injector #2                                                                                |
| E9                                                                                                                               | A    | 10  | #10       | B6 Injector Output 1      | Injector #1                                                                                |
| E9                                                                                                                               | A    | 11  | S1        | MEGA_EPIC_D39             | Shift Solenoid 1                                                                           |
| E9                                                                                                                               | A    | 13  | STA       | MEGA_EPIC_D28             | Cranking Signal                                                                            |
| E9                                                                                                                               | A    | 14  | NSW       | MEGA_EPIC_D32             | Neutral Safety Switch                                                                      |
| E9                                                                                                                               | A    | 15  | IGT3      | B12 Coil 3                | Ignition Coil #3                                                                           |
| E9                                                                                                                               | A    | 16  | IGT2      | B14 Coil 2                | Ignition Coil #2                                                                           |
| E9                                                                                                                               | A    | 17  | S2        | MEGA_EPIC_D41             | Shift Solenoid 2                                                                           |
| E9                                                                                                                               | A    | 22  | RSC       | B18 Low Side Output 2     | Idle Air Control Valve - Close                                                             |
| E9                                                                                                                               | A    | 23  | RSO       | B7 Low Side Output 1      | Idle Air Control Valve - Open                                                              |
| E9                                                                                                                               | A    | 24  | IGT1      | B15 Coil 1                | Ignition Coil #1                                                                           |
| E9                                                                                                                               | A    | 27  | SL        | MEGA_EPIC_D5              | TCC Lockup Solenoid                                                                        |
| E9                                                                                                                               | A    | 30  | OIL       | MEGA_EPIC_ADC_12          | Automatic Transmission Fluid Temperature                                                   |
| E9                                                                                                                               | A    | 31  | PSW       | MEGA_EPIC_D27             | Power Steering Switch                                                                      |
| Connector                                                                                                                        | Plug | Pin | Function  | Tuner Studio Channel Name | Notes                                                                                      |
| E10                                                                                                                              | B    | 7   | VTA (TPS) | D13 TPS1                  | Throttle Position Sensor                                                                   |
| E10                                                                                                                              | B    | 9   | SP2+      | C18 VR1+ Discrete         | Speed Sensor 2 (use selector switch on the board to select between 4Runner/Tacoma or T100) |
| E10                                                                                                                              | B    | 12  | PWR       | MEGA_EPIC_D24             | ECT PWR Button (Automatic Transmission)                                                    |
| E10                                                                                                                              | B    | 13  | OX1       | MEGA_EPIC_A1              | Upstream Narrowband Oxygen Sensor Signal                                                   |
| E10                                                                                                                              | B    | 14  | KNK2      | D14 Knock Input           | Knock Sensor Bank 2 (use selector switch on carrier board to choose KNK1 or KNK2)          |
| E10                                                                                                                              | B    | 15  | KNK1      | D14 Knock Input           | Knock Sensor Bank 1 (use selector switch on carrier board to choose KNK1 or KNK2)          |
| E10                                                                                                                              | B    | 17  | G         | C7 HALL3                  | Cam Signal                                                                                 |
| E10                                                                                                                              | B    | 19  | OX2       | MEGA_EPIC_A15             | Downstream Narrowband Oxygen Sensor Signal                                                 |
| E10                                                                                                                              | B    | 20  | THW       | D16 CLT Coolant           | Engine Coolant Temperature                                                                 |
| E10                                                                                                                              | B    | 21  | THA       | D15 IAT                   | Intake Air Temperature                                                                     |
| Connector                                                                                                                        | Plug | Pin | Function  | Tuner Studio Channel Name | Notes                                                                                      |
| E11                                                                                                                              | C    | 2   | EVP       | B8 LS2                    | EVAP Solenoid                                                                              |
| E11                                                                                                                              | C    | 3   | W         | MEGA_EPIC_D43             | Check Engine Light                                                                         |
| E11                                                                                                                              | C    | 6   | OILW      | MEGA_EPIC_D42             | A/T Fluid Overheat Light                                                                   |
| E11                                                                                                                              | C    | 10  | HT2       | MEGA_EPIC_D12             | Downstream Oxygen Sensor Heater                                                            |
| E11                                                                                                                              | C    | 11  | HT1       | MEGA_EPIC_D13             | Upstream O2 Sensor Heater                                                                  |
| E11                                                                                                                              | C    | 13  | PTNK      | MEGA_EPIC_ADC_14          | EVAP Pressure                                                                              |
| Connector                                                                                                                        | Plug | Pin | Function  | Tuner Studio Channel Name | Notes                                                                                      |
| E12                                                                                                                              | D    | 1   | L         | MEGA_EPIC_D33             | Low Gear Selector (Automatic Transmission)                                                 |
| E12                                                                                                                              | D    | 5   | ACT       | MEGA_EPIC_D49             | A/C Terminate                                                                              |
| E12                                                                                                                              | D    | 6   | OD2/ODLP  | C9 BUTTON 3               | Overdrive Lockout Switch / Light                                                           |
| E12                                                                                                                              | D    | 10  | 2         | MEGA_EPIC_D35             | 2nd Gear Selector (Automatic Transmission)                                                 |
| E12                                                                                                                              | D    | 11  | IDL       | MEGA_EPIC_D40             | Idle Sensor                                                                                |
| E12                                                                                                                              | D    | 12  | SP1       | C5 HALL1                  | Speed Sensor 1                                                                             |
| E12                                                                                                                              | D    | 15  | R         | MEGA_EPIC_D34             | Reverse (Automatic Transmission)                                                           |
| E12                                                                                                                              | D    | 16  | AC1       | MEGA_EPIC_D23             | A/C Request                                                                                |
| E12                                                                                                                              | D    | 22  | ODMS      | C9 BUTTON 3               | Overdrive Lockout Switch / Light                                                           |
| E12                                                                                                                              | D    | 24  | BK        | MEGA_EPIC_D22             | Brake Light                                                                                |